Table 1. Codebook for Web Search Learning: Challenges and Frustrations

Theme	Explanation	Example
Information overload	Overwhelming volume of results; flood of irrelevant pages; too much to filter	"Can be overwhelming to get a large amount of information whose usefulness is unknown without individual evaluation"
Low specificity of results	Results not specific enough to the problem; generic explanations; do not map to the code at hand	"The answers were not always specific enough to my question . . . the blog posts may not directly match the specific problem in the code"
Time-consuming search process	Manually visiting many pages; slow to locate relevant content; paywalls	"Manually visiting different pages to find desired information . . . scrolling through a page to find desired information"
Credibility or freshness concerns	Outdated content; unreliable sources; hard to verify correctness	"We have a lot of authors and some of the information is outdated . . . unable to filter to the specific issues"
Lack of code support or interactivity	No way to input specific code; results require manual mapping to examples	"A lot of the results assumed I had a more developed understanding . . . hard to find examples that reflected the code snippet"
Lack of feedback or confirmation	No indication whether the search direction is correct; no way to verify understanding	"I was not able to find an example that matched the problem, so it was hard to know if I am on the right track"
Requires substantial prior knowledge	Assumes background knowledge; hard to formulate queries without expertise	"I had no real clue about this, so I was searching blindly . . . wasn't sure if the result I got was reasonable"
Query formulation burden	Results depend heavily on phrasing; repeated rewording required	"Using Google forces me to ask better questions, which takes more time to think of"
Difficulty applying information to examples	Concept makes sense in general but unclear how it applies to the given code	"There was lots of information to parse online . . . most resources were not relevant to the specific issue in the code"

Table 2. Codebook for Web Search Learning: Benefits and Strengths

Theme	Explanation	Example
Rich and in-depth information	Detailed articles; in-depth explanations; breadth of available resources	"Reliable sources give a well-detailed explanation . . . variety of platforms with different approaches"
Multiple perspectives and community vetting	Multiple viewpoints from different sources; community-vetted content	"Different sources can be referred to, and the question can be seen in different points of view"
Trustworthy and traceable sources	Identifiable authors; clear source attribution; verifiable references	"It is clear what source information is coming from . . . there is a lot of information easily accessible"
Promotes active thinking and research skills	Encourages independent problem-solving; learner drives the inquiry	"There is more thinking involved . . . more opportunity to come up with your own solution"
Opportunities for broader learning	Encountering adjacent topics while searching; incidental learning beyond the immediate task	"Learning about other things while searching for another"

Continued on next page.

Theme	Explanation	Example
Access to concrete and varied examples	Examples directly available; sometimes more varied than AI-generated responses	"It provided a more in-depth example than ChatGPT and elaborated more, giving multiple examples"
Familiar and easy-to-use interface	Familiar tool; intuitive to query; no concern about prompt phrasing	"It is a simple familiar interface . . . minimal prompts needed"

Table 3. Codebook for LLM Learning: Challenges and Frustrations

Theme	Explanation	Example
Generic or shallow responses	Vague or surface-level explanations; not specific enough to the task	"I was expecting much more in-depth answers . . . the results are in short form and descriptive only"
Single-source limitation	Cannot obtain multiple viewpoints; constrained to what the model chooses to include	"I was limited to one source. When I Googled, I could go to different sites and get different viewpoints and different angles"
Reduced critical engagement	Receiving fully formed answers discourages independent reasoning	"Writes out full answers that doesn't allow me to think critically, since it appears fully formed"
Accuracy or hallucination concerns	Uncertainty about correctness; inability to verify; possible hallucinations	"Unable to verify if the information is correct . . . everything was possibly a hallucination"
Excessive information causing confusion	Long responses; too much at once; hard to extract what is relevant	"The answer was longer and more detailed than what I needed . . . a lot to read"
Cognitive load to interpret responses	Takes time and effort to parse long blocks of text	"Long responses were dumped at once and I had to go back to the beginning . . . hard to parse through"
Overly wordy or unnatural tone	Verbose; formal; repeats itself; unnatural language	"LLMs are very wordy and tend to repeat themselves or speak very unnaturally and too formally"
Prompting skill requirements	Effective prompting requires knowledge the learner may not have; results depend on phrasing	"Having to constantly rewrite my prompts to get the exact output I wanted was a bit frustrating"
Insufficient personalization	Does not adapt to the learner's experience level unless explicitly prompted	"The assisted method didn't tailor its responses to my level of knowledge"

Table 4. Codebook for LLM Learning: Benefits and Strengths

Theme	Explanation	Example
Fast and convenient	Quick answers; time-saving; fewer steps than web search	"It saves time and effort . . . get to the answers quicker than searching through the internet"
Effective problem resolution	Directly identified or resolved the issue; sufficient answer without further searching	"It gave me a correct answer . . . the answer did not contain too much irrelevant content"
Code-specific support	Tailored to the submitted code; line-by-line explanation; directly references the snippet	"It specifically examines my code and finds the problems . . . explained what the problems are and how to solve them"
High relevance to user intent	Specific to the question asked; non-generic output; directly targets the need	"It was specific to my query . . . fast in taking me to the exact line that had the issue"

Continued on next page.

Theme	Explanation	Example
Clear high-level overview	Well-structured summary; organized response; bird's-eye view of the issue	"It gives a good high-level overview . . . the AI assistance resulted in a clear result which identified all the salient points"
Easy to understand explanations	Clear; accessible; beginner-friendly language	"Direct answer to my question . . . easy to understand"
Highly interactive learning	Conversational; allows follow-up questions; dialogue-based clarification	"It was nice to be able to ask follow-up questions when there is confusion about the topic"
Adaptive and personalized assistance	Interprets vague queries; adapts based on follow-up; dynamically adjusts	"ChatGPT made it easier to ask follow-up questions and drill down on what I was looking to understand"
Multiple solution paths	Offers alternative approaches; surfaces possibilities the learner had not considered	"It would give me ideas for further questions to ask . . . gives all the possible scenarios"

Table 5. Codebook for Factors Influencing Participants' Preferences

Theme	Explanation	Example
Efficiency / Time-saving	Preference driven by faster access to answers, fewer steps, or more efficient use of time	"Reduced mental strain . . . quick access to the right information and solutions"
Ease of access / interaction	Method is easier to interact with; conversational or intuitive; less effort to ask or follow up	"Ability to ask follow-up questions for clarity . . . faster and more efficient with less work"
Information clarity / explanation quality	Answers are clearer, more structured, or easier to understand	"Direct, clear explanation . . . I can customize how I wanted it to be explained"
Trustworthiness and accuracy	Confidence or concern regarding factual correctness; reliability of source	"You do not need to check different resources . . . more trust in the generated response"
Depth of learning and support for understanding	Method supports deeper learning; better for conceptual understanding beyond the immediate task	"Traditional search gives you a broader understanding . . . ChatGPT gives you the best answer when in a hurry"
Independence and control	Preference for solving problems independently; avoiding over-reliance on AI	"I prefer searching for my own information. Sometimes what an algorithm thinks I want is not what I actually do"
Specificity and contextual fit	Answers tailored to the user's specific code or situation rather than generic	"Focus on answering my exact question . . . gave sufficient detail at the level that I required"
Other usability-related factors	Familiarity or habit; cognitive load avoidance; enjoyment; balanced or context-dependent preference	"They both have different aspects of usage . . . it depends on the task"